CNN

→ Convolutional Neural Network

Image Segmentation

Healthcare → UNet, UNet++

(CNN)

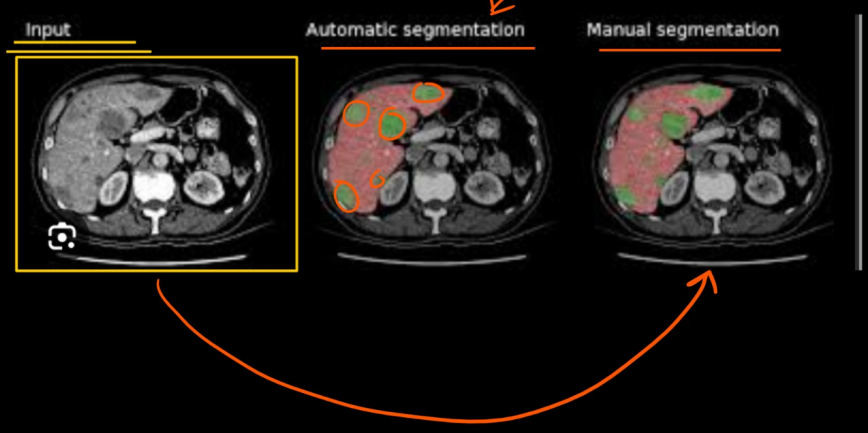


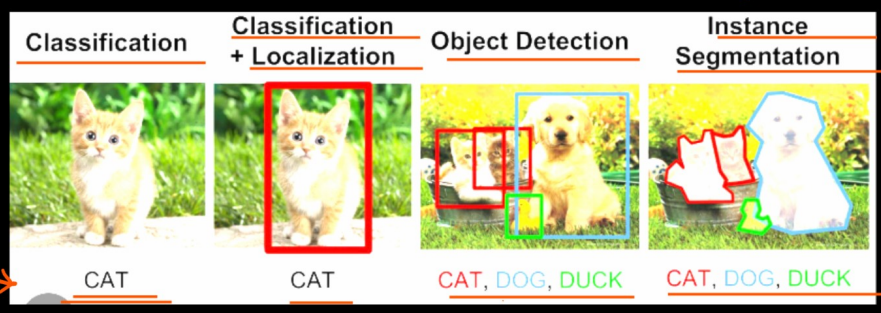
Image Classification

CNN → AlexNet, VGGNet, ResNet

$\boxed{g}$ → g ✓

Object Detection

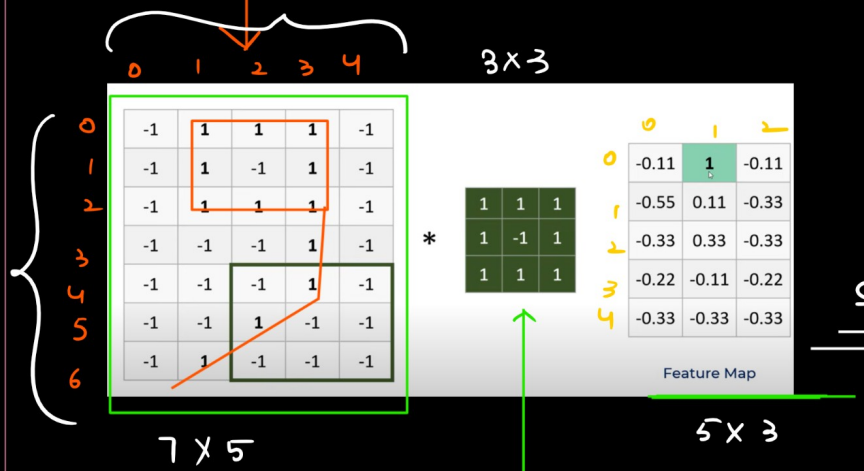
Image



Image

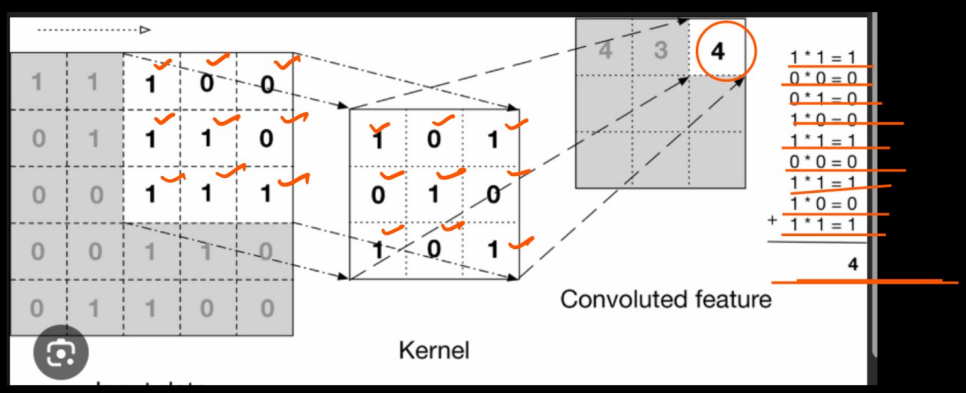
CNN Mame

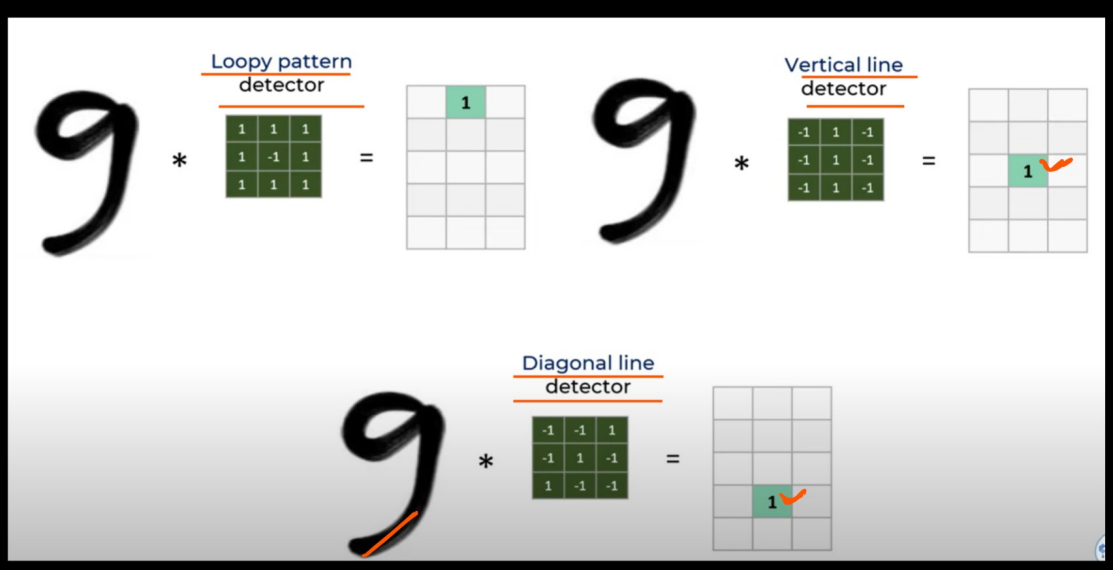
↳ filters → to understand the patterns of the image

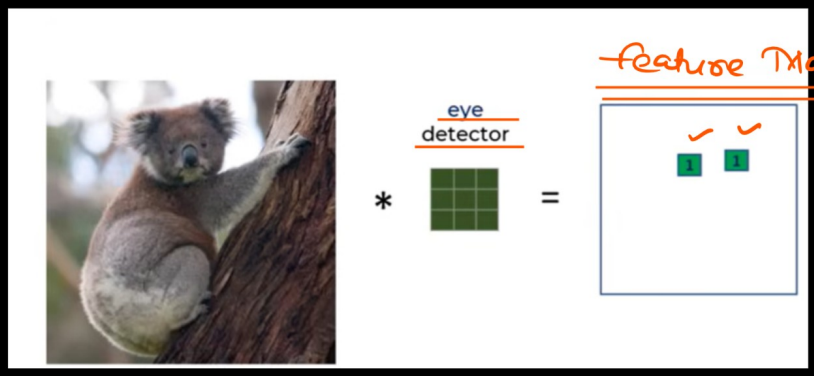


filter / Kernel

Stride = 1







Feature Map

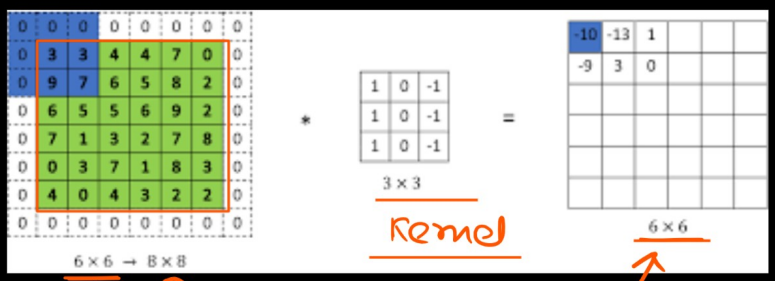
Paddy

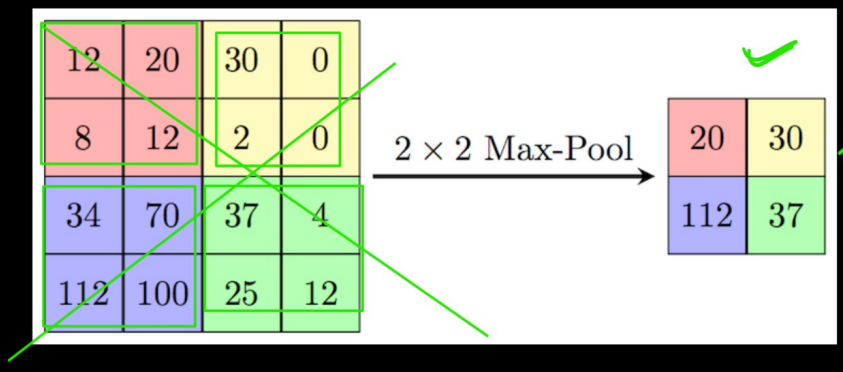
Input shape = 6×6

$$\frac{6 - 3 + 2 * 1}{1} + 1$$

$\longrightarrow$ 6

output shape $\rightarrow$ 6×6





to optimise the Computational cost

Square matrix

$N = 6 \times 6$ (Input image) Padding $= 0$

stride $= 1$

Kernel $= 3 \times 3$

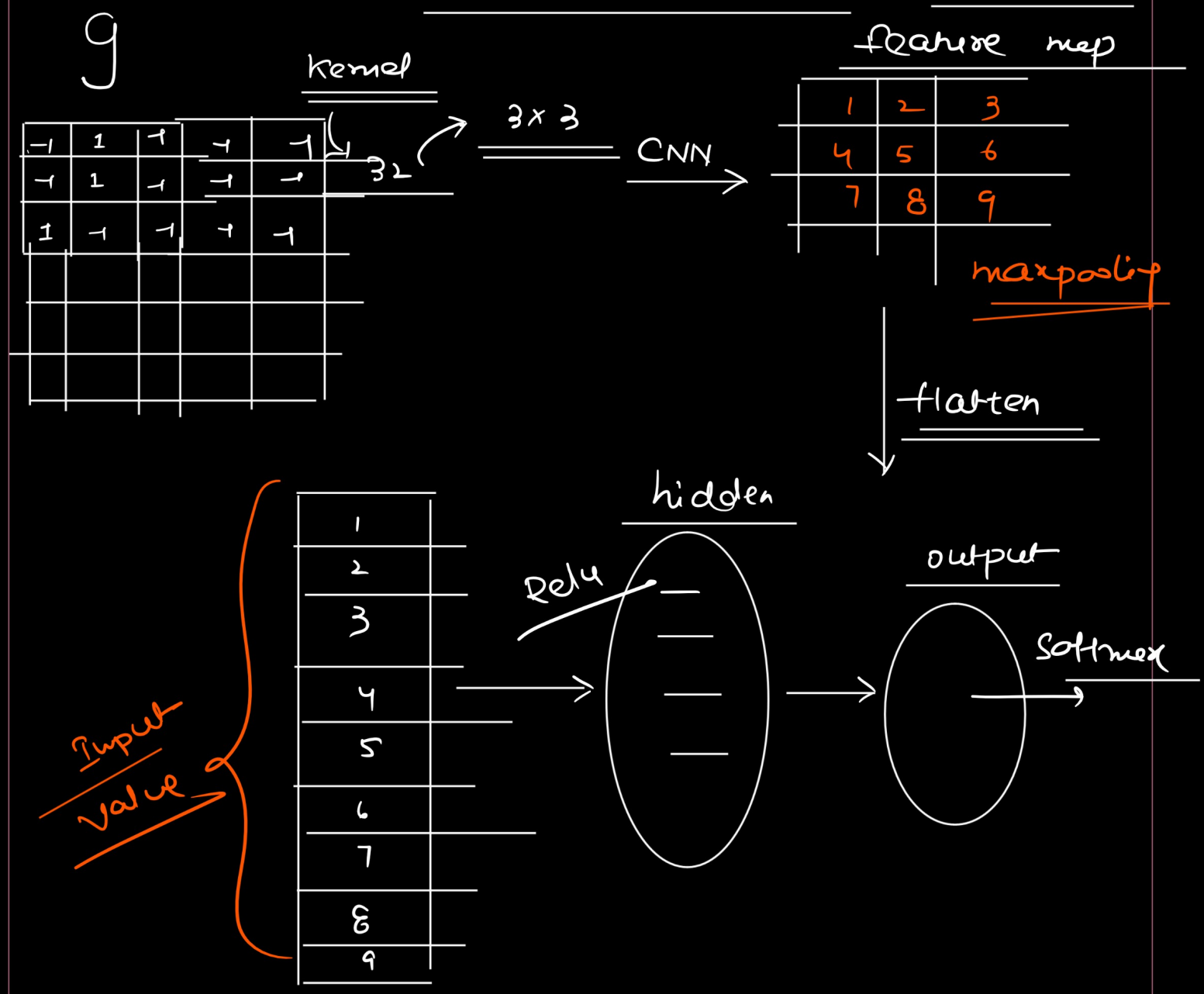
output shape $\Rightarrow$ 4×4

Padding

$$\text{Output feature map} \longrightarrow \frac{N - K + 2P}{S} + 1$$

$$\longrightarrow \frac{6 - 3 + 0}{1} + 1$$

$$\longrightarrow 4 \times 4$$

Neural Network Pipeline

using CNN